MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, MIXER

TYPES 1N21WG, 1N21WGM, AND 1N21WGMR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

• 1.1 Scope. This specification covers the detail requirements for a silicon semiconductor diode types 1N21WG (forward polarity), 1N21WGM (matched forward pair), and 1N21WGMR (matched forward and reverse) for use as a mixer in the S-Band equipment. Ratings for the 1N21WG are applicable to the types having additional suffixes.

• 1.2 Ratings and characteristics.

	F_o	1/G	L	N	VSWR	Burnout
	db	ohms	db	ratio	ratio	erg
Minimum	---	350	---	---	---	---
Maximum	5.5	450	5.5	1.5	1.3	2

OPERATING AMBIENT TEMPERATURE: -65° to +150° C

STORAGE TEMPERATURE: -65° to +150° C

2. APPLICABLE DOCUMENTS

• 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Defense Electronics Supply Center

D64100 - Test Holder for Microwave Diode Type 1N21.

C66058 - Burnout Tester for Microwave Diodes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5961

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THIS DOCUMENT CONTAINS

3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

* 3.3 Design, construction, and dimensions. The semiconductor diode shall be of symmetrical construction and shall consist of a cylindrical body having pins at both ends. The body outline with removable base adapter shall conform to figure 1.

* 3.3.1 Base adapter. The base adapter shall be of the design, construction, and physical dimensions shown in figure 2.

3.3.2 Plating. The diode shall be plated as specified in figure 1.

* 3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

3.5.1 Matched diodes. The 'M' suffix for matched diodes shall be omitted in the type designation on each device. Matched diodes meeting the requirements of this specification shall be packaged with a statement to that effect.

* 3.6 Burnout by single pulse. At the end of manufacturing processes and prior to selecting samples for testing, all diodes shall be subjected to 100-percent burnout by single-pulse test, only once, and this test shall be performed in accordance with Method 4146, MIL-STD-750, using drawing C66058, E = 125 Vdc, minimum, and with the adapter affixed for forward-direction operation.

* 3.7 Measurement error. Allowance for measurement error for electrical tests shall be made in accordance with the accuracy provision of MIL-STD-750, Section 4, and shall be established to the satisfaction of the Government prior to acceptance of the devices by the Government (see 4.5.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Test conditions. Unless otherwise specified herein, the test conditions, when applicable, shall be as follows:

P = 0.5 mW ± 5%

f = 3060 ± 5 MHz

$\overline{F_i}$ = 1.5 db

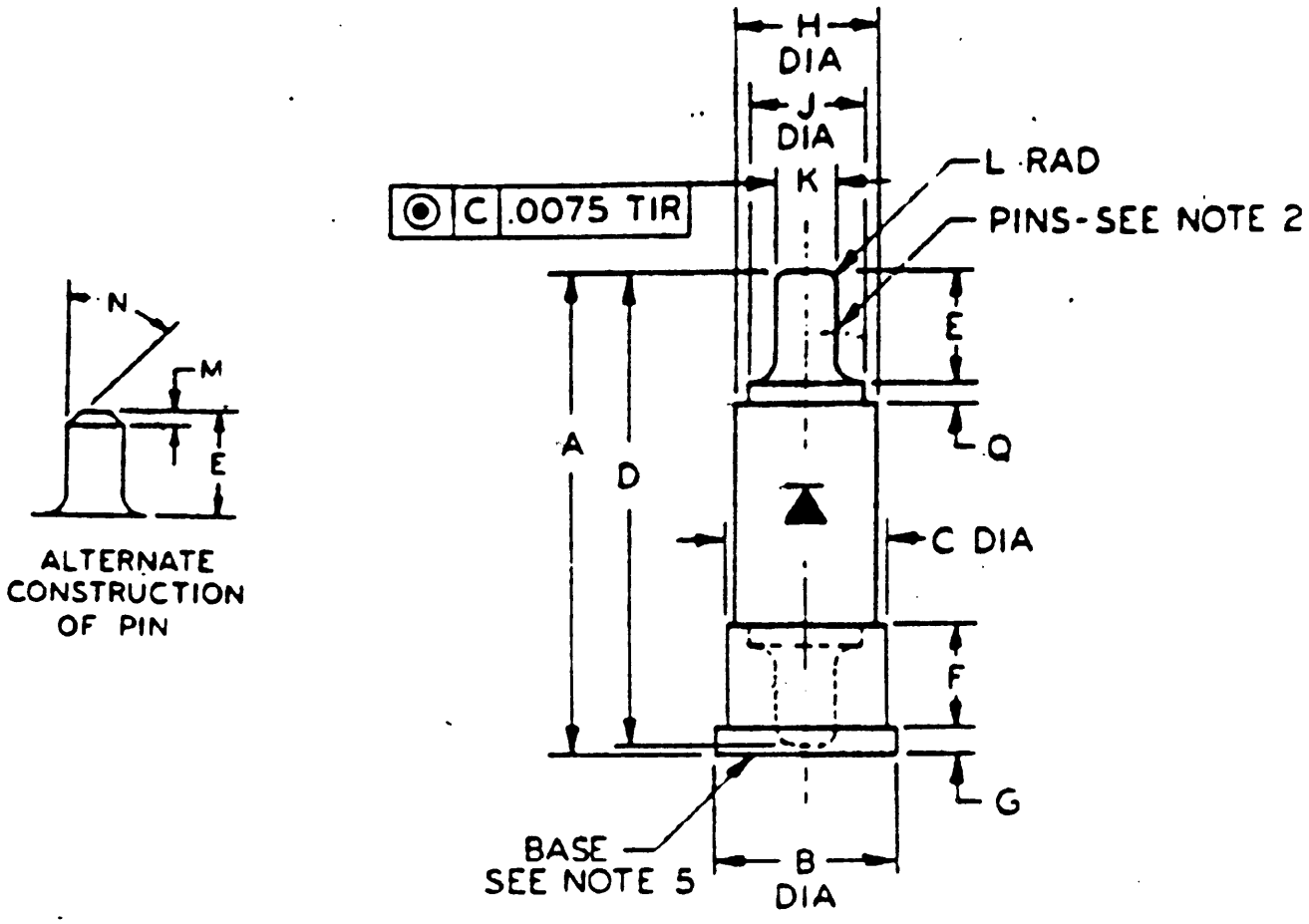
Z_L = 400 ± 10 ohms; ± j0 ohms

R_L = 100 ± 1 ohms

Drawings D64100, C66058

4.2.1 Holder. The diode holder specified in Drawing D64100, or equivalent, shall be used for all electrical test measurements.

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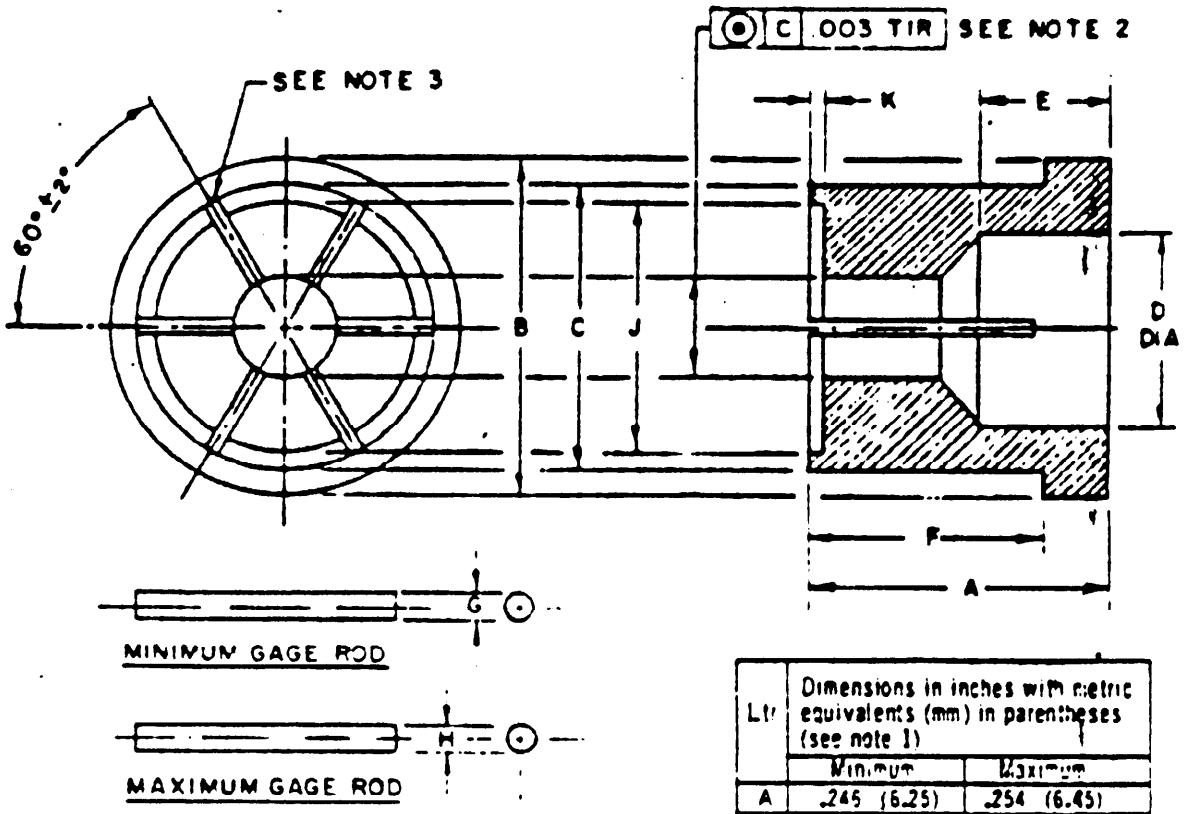


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 1)	
	Minimum	Maximum
A	.800 (20.32)	.840 (21.34)
B	.292 (7.42)	.296 (7.52)
C	.246 (6.25)	.250 (6.35)
D	.766 (19.46)	.792 (20.12)
E	.180 (4.57)	.190 (4.83)
F	.193 (4.90)	.199 (5.05)
G	.047 (1.19)	.057 (1.45)
H	.222 (5.64)	.240 (6.10)
J	.195 (4.95)	.215 (5.46)
K	.092 (2.34)	.094 (2.39)
L	.030 (.76)	.046 (1.17)
M	.010 (.25)	.030 (.76)
N	10°	45°
P	.180 (4.57)	.190 (4.83)
Q	---	.030 (.76)

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Both pins must have convex ends.
3. Metal parts shall be gold plated 10 msi minimum.
4. The polarity shall be indicated by an arrow, with the arrow pointing in the direction of easier current flow.
5. Removable base adapter shall establish the forward or reverse polarity.

FIGURE 1. Semiconductor device, diode, types 1N21WG, 21WGM and 1N21WGMR.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note 1)	
	Minimum	Maximum
A	.245 (6.25)	.254 (6.45)
B	.292 (7.42)	.296 (7.52)
C	.245 (6.25)	.250 (6.35)
D		.187 (4.75)
E		.100 (2.54)
F	.135 (4.95)	.139 (5.05)
G	.0915 (2.32)	.0920 (2.34)
H	.0940 (2.39)	.0945 (2.40)
J	.216 (5.49)	.221 (5.61)
K	.012 (.30)	.018 (.46)

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. This diameter should be .100 (2.54 mm) max diameter before closing jaws. This diameter shall then be sufficiently closed, and the adapter so tempered that it will fit on the minimum gage rod and also on the maximum gage rod, in each case with a snug fit (for hand assembly).
3. Each slot shall be .008 (.20 mm) min, .012 (.30 mm) max wide by .185 (4.70 mm) deep before closing. The 6 slots are equally spaced.
4. The finish for the adapter shall be gold plated 10 mst minimum.
5. All burrs and sharp edges shall be removed.

FIGURE 2. Removable base adapter, included as a part of types 1N21WG, 1N21WGM, and 1N21WGMR

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• 4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III, except the matched-pair requirements in subgroup 3, table I. Qualification testing of both polarities shall be required to obtain qualification approval.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B, and C.

4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

• 4.4.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III.

4.5 Method of examination and test. Methods of examination and test shall be as specified in table I, II, and III, and as follows:

• 4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overall noise figure, shall be raised to +150° C and maintained at this temperature until equilibrium is reached. The $\overline{F}_o$ shall then be determined and shall be less than 10.5 db. The temperature shall then be returned to 25 ± 3° C at which time $\overline{F}_o$ shall be less than the specified limit.

4.5.2 Method of allowing for measurement error. From the stated absolute (true) value limits, corresponding measured value working limits shall be determined by the manufacturer in accordance with the measurement errors expected for a single measurement of each characteristic. For example: A 0.5 db allowance for error in the measurement of $\overline{F}_o$ requires that the working maximum $\overline{F}_o$ limits shall be 0.5 db below the maximum limits stated. This allowance for measurement error shall be made sufficiently large, such that: the probability that the measurement error is less than this allowance is at least 0.90.

4.5.2.1 Determination. The calculations used to determine this allowance shall be based upon the tolerances of the test set components and test conditions, and upon the reproducibility of the readings, in accordance with accepted statistical procedures. These calculations shall be immediately available for, and subject to, Government inspection and revision, and shall be updated when equipment changes or recalibration results in larger errors than originally allowed for.

4.5.3 Matched pair, "M" suffix (forward polarity). The matched forward pair (M suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I, having the anode connected to the adapter.

4.5.4 Matched pair, "MR" suffix (forward and reverse polarity). The matched forward and reverse pair (MR suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I: one diode having the anode connected to the adapter (forward polarity) and the second diode having the cathode connected to the adapter (reverse polarity).

• 4.5.5 Time limit for end points. End point tests for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			7				
Visual and mechanical examination	2071			---	---	---	---

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2			5				
Voltage standing wave ratio (see 4.2)	4136	Dwg D64100 I = 1.0 mAdc		VSWR	---	1.3	ratio
Overall (average) noise figure (see 4.2)	4126	Test cond. A		F_o	---	8.5	db
IF conductance (see 4.2)	4116	Dwg D64100; f = 60 Hz		1/G	350	450	ohms
Subgroup 3			5				
Matched pair requirements: (see 4.5.3 and 4.5.4)							
Conversion loss unbalance	4101			ΔL	---	0.3	db
Intermediate frequency conductance unbalance	4116			$\Delta 1/G$	---	25	ohms

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1							
Physical dimensions	2066	(See figures 1 and 2)					
		Dim. C, J, and K of Fig. 1	7	---	---	---	---
		Dim. A, B, D, E, F, G, H, L, and Q of Fig. 1 (Dim. N, and M and note 3 of Fig. 1 and note 4 of Fig. 2 are for qualification only.)	20	---	---	---	---
Subgroup 2			15				
Thermal shock (temperature cycling)	1051	Test cond. F		---	---	---	---
Terminal strength: Tension	2036	Test cond. A; 1 lb; t = 30 sec		---	---	---	---
Torque	2036	Test cond. D_1; t = 30 sec		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points: (See 4.5.5) Overall (average) noise figure (see 4.2)	4126	Test cond. A		F_o	---	6.5	db

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3			15				
Burnout by repetitive pulsing (forward polarity)	4141	e_o = 15 v, R_g = 50 ohms; 60,000 pulses, min; t_p = 0.05 msec; PRF optional		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			15				
Shock	2016	Nonoperating; 500 G, t = 1.0 msec; 5 blows in each orientation: X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	Nonoperating; 15 G, 50 to 2000 Hz		---	---	---	---
Constant acceleration	2006	Nonoperating; 10,000 G, X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 5			10				
High-temperature operation (see 4.5.1)	---	T_A = 150° C		---	---	---	---
Overall (average) noise figure (see 4.2)	4126	Test cond. A		F_o	---	6.5	db
Subgroup 6			λ = 15				
High temperature life (nonoperating)	1031	T_A = 150° C		---	---	---	---
End points: (Same as subgroup 2)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Conversion loss (see 4.2)	4101	Dwg D64100		L	---	5.5	db
Output noise ratio (see 4.2)	4121	Dwg D64100; R_e = 400 ohms		N	---	1.5	ratio

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REVIEW ACTIVITIES

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500

6. NOTES

6.1 The notes specified in MIL-S-19500 are applicable to this specification.

* 6.2 Superseded diode types. Superseded diode types 1N21G, 1N21GM, and 1N21GMR will be replaced with the 1N21WG, 1N21WGM, and 1N21WGMR respectively. Type 1N21WG, with its removable adapter, may be substituted for the corresponding commercial types 1N21F, 1N21FR, 1N21FM, and 1N21FMR.

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - EL, MI, SM
Navy - AS, OS, MC, CG
Air Force - 14, 19

Preparing activity:
Army - EL

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